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# Bending Responses of The Human Cervical Spine

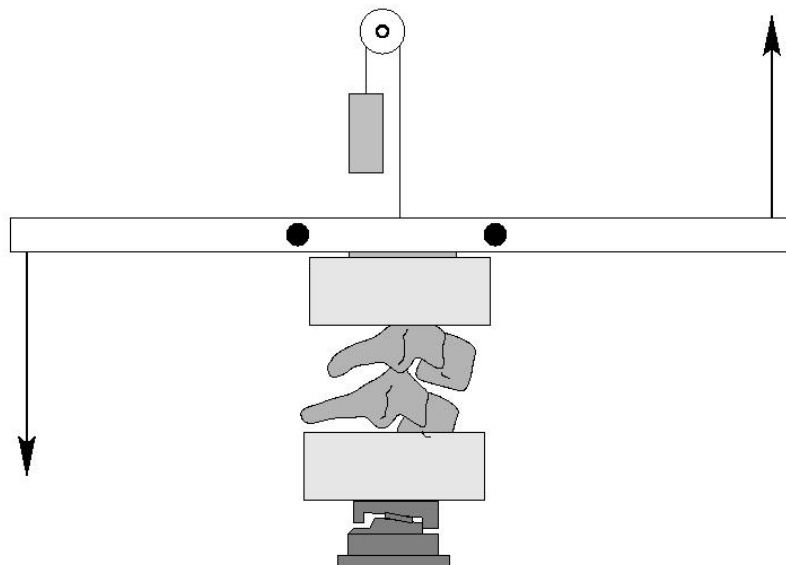
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## Failure Test Report

Test Reference: B16F56FAIL

Test Date: 7/13/99

Submission Date: 9/24/02



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Duke University, Department of Biomedical Engineering  
NHTSA Cooperative Agreement No. *DTNH22-94-Y-07133*

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## Study Objective:

The objective of this study is to provide data on the responses of the human cervical spine to pure bending. The experimental protocol consists of two tests: a flexibility test and a failure test. The flexibility test is designed to quantify the static response of motion segments to flexion and extension moments. The failure test is designed to quantify the bending tolerance of motion segments. See the Methods Section at the end of this report for more details on the experimental protocol.

Specimen ID: B16F

Test Reference: B16F56FAIL

Test Objective: Extension Failure Test

Motion Segment: C5 - C6

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## Specimen Description

Sex: Female Age: 41 years  
Weight: N/A (N) Height: N/A (cm)  
Comments: Osteophyte on C5. Harvest nicks at C6-C7.

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**Injury Summary:** Complete disruption of all ligaments and the disc.

(Y-axis) Moment at Failure: -17.57 N-m  
Angle at Failure: -20.54 °

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## Video Data:

All the digital images are included on the CD which accompanies this report. They are in *tif* format and can be found in the subdirectory, *HSVIDEO*. These represent sequential images, with the frame 0000 corresponding to time 0 of the instrument data. All the angles are computed relative to the specimen Reference Image. The Reference Image is included with the flexibility test report for this specimen.

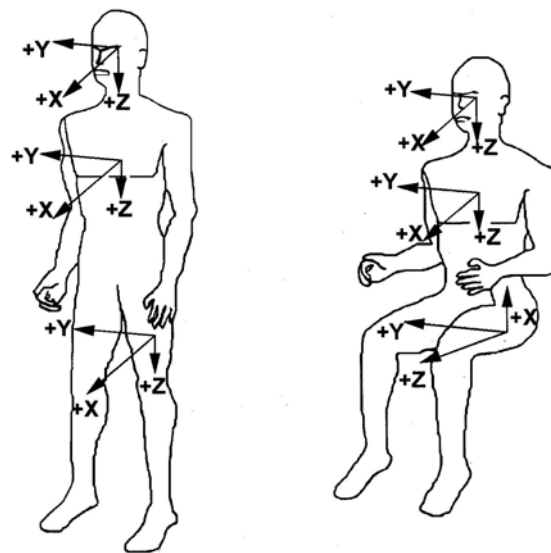
Camera Frame Rate: 125 (frames per second)

**Test Comments:** Cast tilts toward left side.

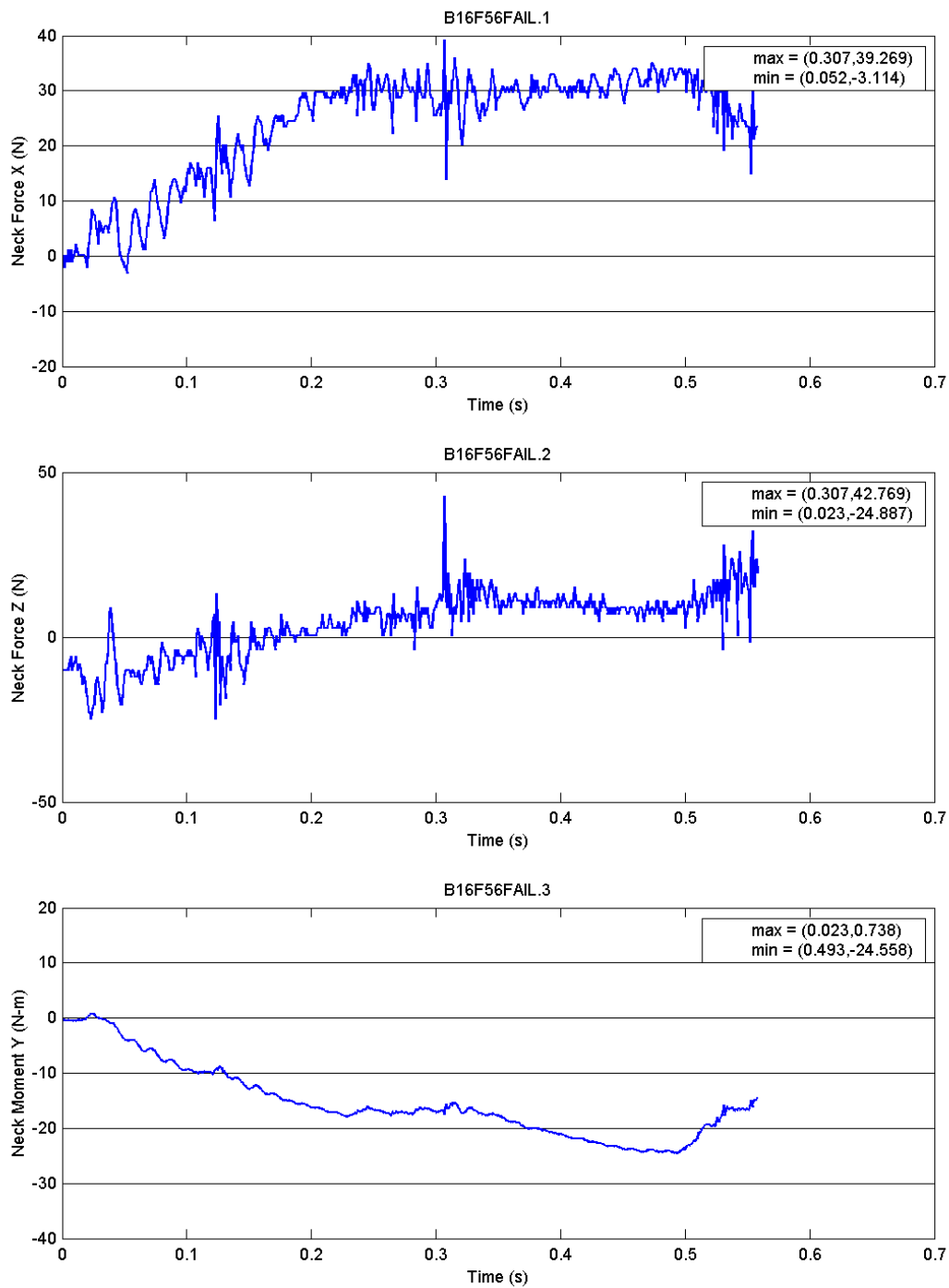
## Appendix A: Coordinate system and sign conventions

Sign Conventions: Compliant with SAE J211/1 March 1995

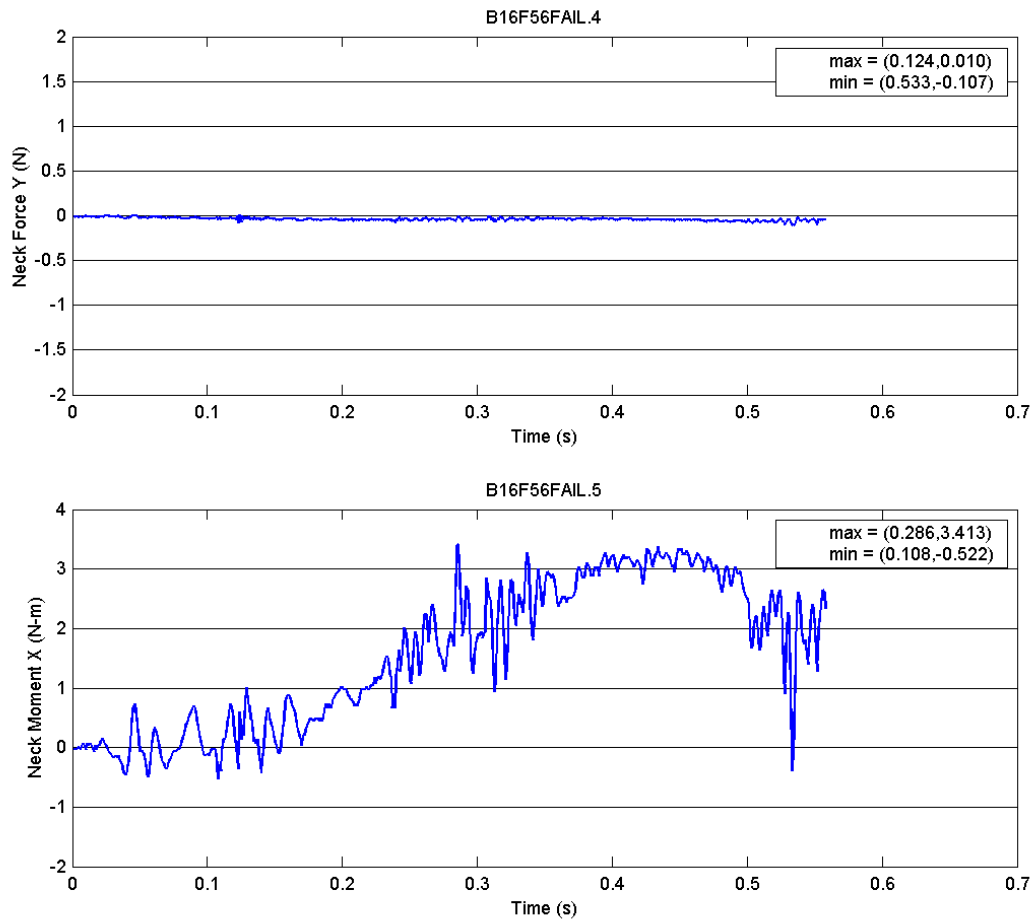
| Load Cell                     | Measure | Manipulation              | Polarity |
|-------------------------------|---------|---------------------------|----------|
| GSE 6607 Upper Neck Load Cell | Fx      | Head Rearward             | +        |
|                               | Fz      | Head Upward               | +        |
|                               | My      | Chin to Sternum           | +        |
|                               | Fy      | Head Leftward             | +        |
|                               | Mx      | Left Ear to Left Shoulder | +        |



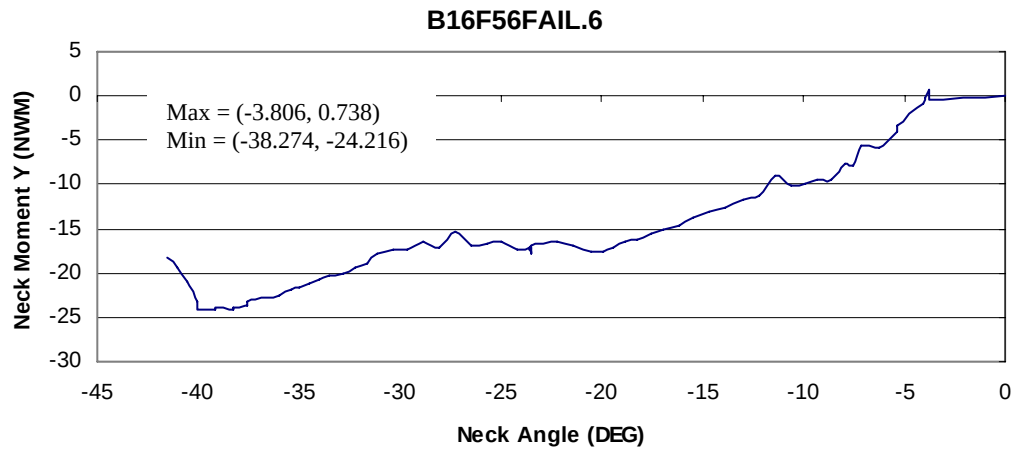
## Appendix B: Instrument Data



## Appendix B: Instrument Data



## Appendix C: Computed Data



## Appendix D: Instrumentation

### Instrumentation

| Channel | Transducer                  | Measure    | Manipulation              | Axis |
|---------|-----------------------------|------------|---------------------------|------|
| 1       | Denton Upper Neck Load Cell | Fx (Force) | Head Rearward             | +x   |
| 2       |                             | Fz (Force) | Head Upward               | +z   |
| 3       |                             | My (N-m)   | Chin to Sternum           | +y   |
| 4       |                             | Fy (Force) | Head Leftward             | +y   |
| 5       |                             | Mx (N-m)   | Left Ear to Left Shoulder | +x   |

### Data Integrity

| Channel | Clip                     | Spike                    | Fail                     |
|---------|--------------------------|--------------------------|--------------------------|
| 1       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 2       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 3       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 4       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 5       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

### Data Acquisition:

Sampling Frequency: **1000 (hz)**

Number of Samples: **559**

### Computed Data:

| Channel | Computation | Measure   | Manipulation    | Axis |
|---------|-------------|-----------|-----------------|------|
| 6       | Flexibility | Angle (°) | Chin to Sternum | +y   |

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## Methods

Unembalmed human heads with intact spines were obtained shortly after death. All specimen handling was performed in compliance with both NHTSA 700-4 guidelines and CDC guidelines. The specimens were sealed in plastic bags, frozen and stored at  $-20^{\circ}$ . All donor medical records were examined to ensure that there were no pre-existing pathologies, such as severe degenerative diseases or spinal pathologies, which could affect the structural responses of the specimens.

The muscular tissues were removed while keeping all the ligamentous structures intact. The lower cervical motion segments (C3-C4, C5-C6, C7-T1) were isolated, cleaned, and cast into aluminum cups with fiber-reinforced polyester resin. The cephalad end of the upper cervical spine specimens was secured using halo fixation. If necessary, the mandible and maxilla were removed in order to allow appropriate flexion range of motion. Upper cervical specimens (Head-C2) were inverted and mounted in the test frame using halo fixation of the head and casting of the C2 vertebra.

The test apparatus applies pure bending moments, and the resulting angular displacements are computed by tracking markers on digital images. The moments are generated using a pair of pneumatic pistons to apply a force-couple. After the specimen was mounted in the test frame, a counterweight was applied to the upper casting assembly. The mass of the counterweight was chosen to counterbalance the mass of the assembly, and to place the motion segment in 0.5 Newtons of tension. The purpose of the 0.5 N tensile load was to establish an initial position for the motion segment in the middle of its neutral zone. The specimen was imaged in this position and the load cell values were zeroed. This image served as the reference position for the flexibility and the failure tests. Prior to flexibility testing, the specimens were preconditioned with 30 cycles of  $\pm 1.5$  N-m of moment. The specimens were then loaded with pure flexion and extension moments in 0.5 N-m increments. Thirty seconds of creep were allowed prior to data acquisition, and the load was released between load steps. The peak applied moment was approximately 3.5 N-m. A six-axis load cell (GSE model 6607-00) at the base of the specimen was used to ensure that the moment remained pure. The load cell is also the source of the moment data.

After completion of the above bending tests, the tension mass was removed, and the specimens were failed in either flexion or extension. The loading rate was the maximum that could be achieved using our 100 psi system. The rates were dependent on the stiffness of the specimen, and were on the order of 90 N-m/second. The failure tests were imaged at 125 frames/second. Failure was defined as a decrease in the measured moment with increasing rotation, and they were verified by examination of the high speed images. Specimen dissection was performed to document injuries.

All the image data was uploaded to an SGI running a modified version of the xv shareware package. This software enables us to digitize the location of the marker pins in each image. The location of each marker is determined using a centroid algorithm to increase accuracy. Angular displacements for each image are determined with respect to the reference image.